A Certified Canonical Census of One-Place Stark Invariants over Real Quadratic Fields

Exact Quadratic Value Orbits and the Higher-Order Frontier

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Abstract

We give a certified finite census of canonically selected one-place differenced Stark packets over real quadratic fields. For squarefree radicands $2 \leq D \leq 200$ and finite-ideal norm at most 100, we first quotient finite ideals by field conjugation, choose the least normalized HNF in each orbit, and then attach the pinned real place ∞_2. This gives 8,200 selected moduli, which split exactly into 3,936 empty-support rows, 1,560 quadratic-support rows, and 2,704 higher-order rows. Every quadratic row has an exact squarefree polynomial for its distinct packet-value orbit over the real quadratic base. A deleted-prime cover criterion decides, before any regulator or unit computation, exactly when the whole quadratic packet is one. The higher-order table has complete registered-mechanism status for 2,699 rows and preserves five tool-incomplete quartic rows. These are statements about the selected finite universe only: we claim neither a census of all pointed one-place moduli, an asymptotic density, nor a new higher-order packet identity.

1 Claim boundary and conventions

Let K be real quadratic and $\mathfrak{m} = \mathfrak{f}\infty_2$ a one-place modulus. With R the real-place sign class, put

$$Z_{\mathfrak{m}}(s, A) = \zeta_{\mathfrak{m}}(s, A) - \zeta_{\mathfrak{m}}(s, RA), \qquad X_A = \exp Z'_{\mathfrak{m}}(0, A).$$

Here $\mathrm{Cl}_{\mathfrak{m}}(K)$ is the ray class group for the indicated finite and infinite modulus. The packet is the Artin-indexed tuple $(X_A)_{A\in\mathrm{Cl}_{\mathfrak{m}}(K)}$. The exact Fourier, Artin-action, place-ordering, and positive-orientation conventions are those of the companion results paper [1]; this paper does not introduce a second convention system.

The proved assertions here are the finite trichotomy, the exhaustive quadratic packet-value-orbit corpus, the quadratic deleted-prime cover criterion, and twelve named higher-order member transports. The higher-order eligibility table is an exact finite-population observation: five quartic rows remain incomplete. We do not assert results beyond the frozen selected range. Eligibility is not a packet identity. In particular, a common normal closure or a banked representative does not promote another member: exact conductor, ray-class, orientation, and Artin-labelled packet transport are separate obligations. The registered Shintani-transfer scope has 232 eligible rows. Its successor ledger records twelve completed noncanonical member transports; the other 220 remain open.

We use *PROVED* when a cited theorem applies after exact checks and *OBSERVED* for reproducible descriptive statistics of this finite population. The tag `CERTIFIED_NUMERICAL` denotes a rigorous enclosure with a printed radius criterion. "Unconditional" means that an asserted identity follows from a proved theorem after exact hypothesis checks; it does not prove the general Stark conjecture.

2 Frozen universe and trichotomy

The selected universe consists of maximal-order fields $\mathbb{Q}(\sqrt{D})$ for squarefree $2 \le D \le 200$ and nonzero integral ideals $\mathfrak{f}$ with $\mathrm{N}\,\mathfrak{f} \le 100$. In the frozen integral basis, replace $\mathfrak{f}$ by the lexicographically smaller normalized HNF of $\mathfrak{f}$ and $\bar{\mathfrak{f}}$, then attach the pinned place ∞_2. Thus conjugation is applied to the finite ideal before the distinguished place is fixed. This is a canonical selected-modulus census; it is not the full isomorphism-class quotient of all pairs $(\mathfrak{f}, \infty_i)$.

A clean PARI replay enumerates 121 fields and 13,939 raw ideals. Of these, 2,461 are self-conjugate and 11,478 are nonself-conjugate, so the implemented finite-ideal quotient has

$$2461 + \frac{11478}{2} = 8200 \tag{1}$$

representatives. No theorem below promotes this selected universe to a census of all pointed one-place moduli.

Let T be empty differenced Fourier support; let Q be nonempty support in which every supported character has order two; and let H be nonempty support containing a character of order greater than two. The structural sign-class lemmas in [1] show that empty support implies $X_A = 1$ for every A. The converse is false: a nonempty supported term can be annihilated by an imprimitive Euler factor.

Theorem 1 (Frozen trichotomy). *In the frozen universe,*

$$|T| = 3936, \qquad |Q| = 1560, \qquad |H| = 2704.$$

Every row belongs to exactly one stratum. Every packet in T is one.

Proof. The clean enumerator certifies every quadratic base field and applies the canonicalization in (1). Exact character inversion computes the differenced support of every retained row. The three predicates above are disjoint and exhaustive; their audited counts are 3936, 1560, and 2704. If the support is empty, Fourier inversion gives $Z_{\mathfrak{m}}(s, A) = 0$ identically, hence $X_A = 1$. $\square$

The earlier v5 routing declaration recorded 3,899 trivial rows. The remaining 37 are empty-support rows that its obsolete routing order assigned an exponent-cap label before applying the empty-support theorem. The historical record is preserved; Theorem 1 uses the support-first replay.

The set T is therefore a structural empty-support stratum, not the set of all rows whose evaluated packet is one. In particular, the quadratic stratum below contains a disjoint 346-row value-one degeneracy sub-stratum.

3 The exact quadratic stratum

For a supported quadratic character, the uniform theorem in [1], based on Tate's rank-one algebraicity theorem [2, Thm. IV.5.4], gives exact exponents from imprimitive Euler factors and a relative regulator index. Conductors, Euler factors, relative-unit kernels, orientations, and indices are checked exactly. There are 2,232 supported character occurrences: 672 have zero Euler factor, and 346 rows have all factors zero and packet polynomial $X - 1$.

More explicitly, for $G = \mathrm{Cl}_{\mathfrak{m}}(K)$ the theorem gives

$$X_A = \prod_{\chi(R)=-1} |\tau_1 u_\chi|^{c_{A,\chi}}, \qquad c_{A,\chi} = \frac{4}{|G|}\chi(A)^{-1} E_\chi \frac{h_{L_\chi}}{h_K}\frac{w_K}{w_{L_\chi}}\frac{1}{I_\chi}. \tag{2}$$

Every quantity in the exponent and every oriented relative unit is computed by exact arithmetic.

If u_i is a norm-one relative unit, then

$$t_i = u_i + u_i^{-1} = \mathrm{Tr}_{L_i/K}(u_i) \in K.$$

The formal sign-orbit polynomial is therefore synthesized without its compositum by

$$P_0(x) = x - 1, \qquad P_i(x) = \mathrm{Res}_z\big(P_{i-1}(z), x^2 - t_i x z + z^2\big). \tag{3}$$

Each root z is replaced by zu_i and zu_i^{-1}, so every intermediate polynomial remains over K. The recurrence concerns denominator-cleared packet powers. At denominator two, the positive packet factor is selected only after exact square-lift, reciprocity, irreducibility, positivity, and Artin-image checks.

For a quadratic row define its *packet-value-orbit polynomial* by

$$F_{\mathfrak{m}}(x) = \prod_{v \in \{X_A : A \in G\}_{\text{distinct}}} (x - v).$$

This squarefree polynomial records the distinct effective Artin orbit; it is not, by itself, a class-to-root labelling. The exact ray-group coordinates, sign-class log, supported characters, and exponents in (2) are retained so that the Artin-labelled tuple can be reconstructed on replay. Only the worked row below prints that full labelling in the present article.

Theorem 2 (Exhaustive quadratic stratum)**.** *Every one of the 1,560 rows of Q has a certified exact squarefree packet-value-orbit polynomial $F_{\mathfrak{m}}(x)$ over its base field. Its degree equals the exact effective Artin-image size. The stable RQ-ordered row certificates have terminal hash-chain root*

7c04242b1d4c11293af96f83f4915dbed25f6125c60d82965e533df5c9d81855.

Proof. Apply (2) character by character. Clear the common rational denominator, synthesize the formal sign orbit by (3), and restrict it to the exact Artin sign image. For the 69 denominator-two rows, exact squareness in the full ray field selects the positive lift. Each retained factor is then checked for reciprocity, squarefreeness, irreducibility over K, positive real-root signature, and equality of its degree with the effective Artin-image size. All 1,560 row checks pass. The hash chain protects the ordered corpus against later alteration; it is an integrity record, not a substitute for these algebraic checks. □

For a supported quadratic character χ, let χ° be its primitive character, let $\mathfrak{f}_\chi$ be its finite conductor, and put

$$\mathcal{D}_\chi = \{\mathfrak{p} : \mathfrak{p} \mid \mathfrak{f},\ \mathfrak{p} \nmid \mathfrak{f}_\chi\}.$$

These are the prime ideals deleted when the primitive L-function is made imprimitive at $\mathfrak{f}$.

Theorem 3 (Deleted-prime cover criterion)**.** *In the quadratic-support regime, the following are equivalent:*

1. $X_A = 1$ *for every* $A \in G$;
2. $L'_{\mathfrak{m}}(0, \chi) = 0$ *for every* $\chi(R) = -1$;
3. $E_\chi = 0$ *for every* $\chi(R) = -1$;
4. *for every* $\chi(R) = -1$ *there is a* $\mathfrak{p} \in \mathcal{D}_\chi$ *with* $\chi^\circ(\mathfrak{p}) = 1$.

Equivalently, the deleted primes that split in the quadratic extension cut out by χ° cover every supported character.

Proof. The imprimitive factor is

$$E_\chi = \prod_{\mathfrak{p}\in\mathcal{D}_\chi} (1 - \chi^\circ(\mathfrak{p})).$$

Every displayed local value is 1 or -1, because χ° is quadratic and unramified at the deleted prime. Hence $E_\chi = 0$ exactly when one such value is 1, proving the equivalence of (3) and (4). The class numbers, roots-of-unity quotient, index, and oriented log $|\tau_1 u_\chi|$ in (2) are nonzero, so the explicit quadratic formula gives (2) if and only if (3). Fourier inversion gives (2) implies (1). Conversely, (1) gives $Z'_{\mathfrak{m}}(0, A) = 0$ for every A; character orthogonality applied to the exact inverse transform recovers (2). □

The criterion is independent of the census bounds and is an exact pre-regulator decision test. In the frozen range its deleted-prime export checks 2,232 supported character occurrences and 1,516 deleted prime occurrences. Exactly 699 deleted occurrences have local value 1, and the resulting cover holds on exactly 346 rows: 296 with one supported character and 50 with two. It has zero character-level and row-level classification errors, and agrees with $F_{\mathfrak{m}} = x - 1$ on all 1,560 exact packet polynomials.

No four-support row in the frozen range is covered, but this is not a general theorem. A preregistered falsification search finds the exact counterexample $K = \mathbb{Q}(\sqrt{7})$ and finite-ideal HNF $[42, 0; 0, 6]$ of norm 252. Its ray group is $C_6 \times C_2 \times C_2$, its sign log is $(3, 1, 0)$, and its four supported characters all have a deleted prime with local value 1. Thus all four Euler products vanish and the packet is one. This row lies outside the frozen census and refutes the tempting universal "four-support nondegeneracy" pattern.

Table 1: Exact quadratic-stratum distributions.

invariant	count
packet degree $1, 2, 4, 8$ over K	$346, 930, 242, 42$
common denominator $1, 2$	$1491, 69$
all-Euler-zero packet $X - 1$	346
maximum coefficient-coordinate size	62 digits

Here the coefficient-coordinate size is the maximum decimal digit count of the numerator or denominator of either coordinate in the frozen basis $1, y$ of K. The preregistered height-only predictor had maximum 89 digits, below the frozen 256-digit cap. The full exact run took 43.46 seconds under PARI/GP 2.15.4; this is an OBSERVED engineering statistic, not a portable benchmark.

A separately frozen 50-row audit used the independent analytic class-number/regulator formula before opening the corpus traces. At 384 bits all 43 nonzero character comparisons and 101 Artin sign rows pass the preregistered 10^{-38} difference-radius cap; the independent balls were inflated by 10^{-45}. This is a CERTIFIED_NUMERICAL consistency check, not a proof input. The initial 192-bit run failed the radius gate at RQ-006617; that failure is preserved rather than discarded.

3.1 A worked imprimitive row

RQ-000013 over $\mathbb{Q}(\sqrt{2})$ has finite ideal HNF $[14, 6; 0, 2]$ of norm 28 and ray group C_2. Its supported character has primitive conductor obtained by removing one prime of norm two. The exact imprimitive Euler factor and relative index are both two:

$$E_\chi = 2, \qquad I_\chi = 2.$$

The oriented relative unit has minimal polynomial

$$u^4 - 2u^3 + u^2 - 2u + 1$$

and its selected real root lies in $(9/5, 19/10)$. Exact Artin labels then give

$$X_{[0]} = u^2, \qquad X_{[1]} = u^{-2}.$$

This is a PROVED application of the uniform theorem, including the otherwise easy-to-miss nonzero imprimitive Euler branch. A numerical point-value comparison is retained only as an OBSERVED cross-check.

4 Higher-order taxonomy and frontier

For every H row we record the status of three registered mechanisms. The Shintani-transfer column requires the exact operational predicates used in the companion theorem: derived-subgroup order two, the real-place and unit-congruence gates, and the frozen effective-exponent bound. The cyclic-quartic CM column requires order-four nonquadratic support, projective V_4 geometry, exact local factors, the registered S-set condition, roots of unity, and orientation. These two columns record route eligibility, not packet identities.

The Roblot column tests Theorem 7.1 [7]: conditions (A1)–(A3), equality with the theorem's S-set, class number prime to three, and absence of wild ramification above three. Its conclusion is the theorem's weak cyclic-sextic statement up to absolute values, not an Artin-labelled positive packet. The registered-mechanism status is complete on 2,699 rows; five old quartic constructions remain tool-incomplete. Every sextic applicability decision is complete.

Table 2: Route-by-Roblot cross-table for the finite H stratum.

registered route	weak coverage	hyp. fail	unsupported	incomplete	total
Shintani transfer	70	45	117	0	232
cyclic-quartic CM	782	99	0	0	881
exclusive frontier	227	261	1098	5	1591
total	1079	405	1215	5	2704

Thus 1,359 rows—the 261 hypothesis failures and 1,098 unsupported orders in the exclusive frontier—fail every completed registered mechanism. The 227 frontier rows with Roblot weak coverage are not Artin-labelled packet identifications, and the five incomplete rows are not counted as failures.

For order-six support, 764 character/inverse-character kernels reduce to 407 primitive-field keys. Exact local checks remove 25 keys and 382 require field certificates. The residual 73 possible three-divisible class-number fields are each closed by an exact unramified cyclic-cubic certificate. Thus every sextic kernel has an exact Roblot applicability decision. The five incomplete entries of Table 2 are old quartic construction failures, not missing sextic decisions and not mathematical no-go results.

The index column uses the genuine derived-subgroup order in the common stable modulus, rather than the legacy W1 screening proxy. This distinction resolves a potentially misleading control case: RQ-005298 has legacy displayed index two, but its certified normal closure has relative degree 128, maximal absolutely abelian subfield of relative degree 32, and therefore derived-subgroup order $128/32 = 4$. It fails the actual Engine-B index-two predicate; its non-eligibility is not an unstated failure of a closure, conductor, or transfer condition. The separately incomplete quartic Engine-C resolvent for this row has no mathematical verdict.

The headline wall is RQ-000692 over $\mathbb{Q}(\sqrt{21})$, with finite norm 36 and support profile $(2, 6)$. Its Shintani index is six. The one-place, local-S, and prime-to-three-class-number conditions of Roblot's theorem pass, but a prime above 3 is wildly ramified in the relevant sextic extension. The theorem therefore does not apply. This is an exact limitation of the registered methods, not a non-algebraicity theorem.

4.1 Minimal frontier targets

The frontier artifact selects the least RQ identifier at each support order, independently of the outcome of later mechanism checks. A representative initial segment is given in Table 3. The complete table, including orders through 210, is machine-readable in the H-taxonomy artifact. The distinction between "unresolved" and "all mechanisms fail" is retained: an unresolved row can still be eligible for one of the registered mechanisms.

Table 3: First frozen frontier targets by support order.

order	minimal unresolved	minimal all-mechanisms-fail
4	RQ-000030	RQ-000277
6	RQ-000032	RQ-000227
8	RQ-000018	RQ-000018
10	RQ-000078	RQ-000078
12	RQ-000294	RQ-000294
14	RQ-000524	RQ-000524
16	RQ-001056	RQ-001056
18	RQ-000657	RQ-000657
20	RQ-000785	RQ-000785
24	RQ-000363	RQ-000363

5 Engine-B transport scope

The registered Shintani-transfer population consists of 232 rows in 88 groups by exact normal-closure polynomial. A successor ledger promotes exactly the twelve noncanonical members in Table 4. For RQ-000039, for example,

$$X_{98}(A) = X_{49}(A)X_{49}(A\mathfrak{q}^{-1})^{-1},$$

where $\mathfrak{q}$ is the unique norm-two prime of $\mathbb{Q}(\sqrt{2})$. Every row uses exact Euler deletion and a positive product or quotient at the frozen split embedding; this is a distribution relation, not equality of packets. In the table, c means that target label a maps to source label ca in C_6, and (ℓ, e) records a deleted prime with source ray log ℓ and quotient exponent e.

The B5-086 direct-source closure shares source prime 11 in every target enlargement, while B5-021 and B5-033 admit no integral source-to-target modulus quotient. These are exact exclusions from the coprime Euler-deletion route, not negative packet claims. The remaining 220 members retain status `UNSTARTED_NO_CASE_LEVEL_PACKET_CLAIM`.

Six historical rows—RQ-000970, RQ-000991, RQ-000993, RQ-001004, RQ-002416, and RQ-004845—have a different printed normal-closure polynomial in the reconstruction. Both records are retained. Literal polynomial equality is not used to infer field identity or packet transport, and none of these discrepancies is used as a case-level identity. This is why mechanism eligibility remains separate from higher-order packet proof.

Table 4: All proved noncanonical Engine-B member transports.

target	source	closure	c	deleted (ℓ, e)	status
RQ-000039	RQ-000021	B5-015	1	$(1,1)$	PROVED
RQ-000195	RQ-000190	B5-025	1	$(1,1)$	PROVED
RQ-000200	RQ-000190	B5-025	1	$(2,1)$	PROVED
RQ-000205	RQ-000190	B5-025	1	$(1,2)$	PROVED
RQ-000213	RQ-000190	B5-025	1	$(1,1),(2,1)$	PROVED
RQ-000221	RQ-000190	B5-025	5	$(1,3)$	PROVED
RQ-000228	RQ-000190	B5-025	5	$(5,2)$	PROVED
RQ-000425	RQ-000419	B5-022	1	$(4,1)$	PROVED
RQ-000436	RQ-000419	B5-022	5	$(1,1)$	PROVED
RQ-000457	RQ-000419	B5-022	5	$(4,1),(4,1)$	PROVED
RQ-000459	RQ-000419	B5-022	1	$(5,1)$	PROVED
RQ-000465	RQ-000419	B5-022	5	$(3,1)$	PROVED

6 Reproducibility and relation to earlier work

Every material assertion has a versioned JSON artifact, input hashes, and replay code under PARI/GP 2.15.4. From the project root, the local revision gates are

`python3 scripts/audit_census_paper.py` and `python3 scripts/audit_census_referee_revision.py`.

The quadratic theorem gate is `python3 proof/audit_q_euler_deleted_prime_cover.py`. The primary finite evidence is the support-first reconciliation, the row-level quadratic corpus, the deleted-prime cover certificate, and the H-taxonomy matrix. The deterministic paper-and-replay package is archived at `https://doi.org/10.5281/zenodo.21729947`. The companion paper proves the uniform quadratic theorem and selected higher-order identities; this census does not reprove or enlarge them.

Table 5 states the exact comparison boundary. No row count is assigned to a prior theorem without a row-level map of its extension, S-set, distinguished place, normalization, and conclusion.

Table 5: Prior-theorem boundary used in this census.

source	theorem/object	use and remaining boundary
Tate [2]	Thm. IV.5.4, rank one	Algebraicity input for the quadratic formula after exact conductor, Euler, class-number, unit-index, and orientation checks.
Dummit–Sands–Tangedal [5]	Thms. 2 and 4	Refined multiquadratic and full biquadratic antecedents; no census-row coverage count is claimed without the still-open row-level hypothesis map.
Roblot [7]	Thms. 5.5, 6.1, 7.1	Quadratic, cyclic quartic, and conditional cyclic-sextic weak results. Table 2 counts only exact Theorem 7.1 weak coverage, never Artin-labelled packet identification.
Cohen–Roblot [6]	Hilbert class-field algorithm	The objects differ from one-place ray moduli. Four exact subfield controls test field containment only, not unit or packet overlap.

Arakawa's relative-index formula [3] and the explicit computational work of Dummit–Sands–Tangedal [4] are further antecedents, but neither is assigned a census-row count here.

The prior-art comparison is deliberately narrow. Cohen–Roblot study Hilbert class fields, whereas the present promoted examples and census rows are one-place ray-field objects with nontrivial finite modulus. An exact four-control subfield audit is mixed: the selected ray normal closures for bases 35, 42, and 186 contain the associated Hilbert biquadratic field, while the selected base-51 closure does not. This is a PROVED statement about the four frozen normal closures only; it identifies neither a Stark unit nor an Artin-labelled packet and does not generalize containment. This paper makes no priority assertion from any of these comparisons. The contribution claimed here is the canonical finite classification and the exhaustive quadratic packet-value-orbit corpus within its corrected boundary.

A Record map and replay boundary

The trichotomy is in `artifacts/census-paper-layer0-reconciliation-v1.json`; the quadratic audit is in `artifacts/census-q-packet-corpus-audit-v1.json`; the H matrix is in `artifacts/census-h-taxonomy-v2.json`; the range-description correction is in `data/census-paper-preregistration-amendment-v18.json`; the preserved v1 record used the legacy displayed-index proxy and is corrected by the v2 provenance fields. The quadratic deleted-prime theorem and its expanded-range counterexample are in `artifacts/q-euler-deleted-prime-cover-theorem-v1.json`. Engine-B transport scope is in `artifacts/engine-b-transport-manifest-v5.json`; its successor ledger is in `artifacts/engine-b-transport-ledger-v4.json`. The latter records exactly twelve completed member transports. The exact direct-source coprime no-go screens are recorded in `artifacts/b5086-transport-geometry-v1.json` and `artifacts/final-direct-source-coprime-screen-v1.json`. The exact four-control Hilbert/ray object audit is in `artifacts/b5079-hilbert-ray-containment-v1.json` and `artifacts/hilbert-ray-containment-tranche-v1.json`; it is field containment only. The versioned Cycle-127 correction record freezes the revised source hashes, audit results, and replay resource use. The release record, including the root-file inventory and public checksums, is associated with DOI `https://doi.org/10.5281/zenodo.21729947`.

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